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Project Based Learning Activity

Course Name: Database Management System

Project Title

Hospital Management System

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Contents:

1. Abstract

This project details the design and implementation of a comprehensive, secure **Hospital Management System (HMS)** built on a modern **Full-Stack Architecture** (Node.js/Express, React, and MySQL). The system addresses key operational inefficiencies within contemporary healthcare environments by digitizing critical workflows and establishing rigorous role-based access control (RBAC). The HMS features integrated portals for three primary user roles: **Patients**, **Doctors**, and **Receptionists**. Key functionalities include **Bcrypt-secured authentication**, real-time appointment scheduling and cancellation, inventory management for pharmaceuticals, and advanced **Medical Record Management** that supports the secure upload and linking of external files (e.g., X-rays) using the Multer middleware. By establishing a single source of truth for all patient and administrative data, this application significantly enhances workflow efficiency, ensures data integrity via SQL schema enforcement, and provides a scalable, professional foundation for digital healthcare operations..

2. Keywords

Hospital Management System, Full-Stack, Node.js, React, MySQL, Role-Based Access Control, Bcrypt, File Uploads, Database Normalization..

3. Introduction

The rapid modernization of healthcare mandates a transition from paper-based record-keeping to unified electronic systems. Traditional methods are prone to administrative error, data duplication, and security vulnerabilities, particularly in multi-specialty environments. This project is motivated by the need to develop a robust, integrated HMS capable of managing patient lifecycles, from initial appointment scheduling and secure login to comprehensive medical documentation and billing. The system is designed to demonstrate key database and architectural principles, proving the viability of a custom full-stack solution for modern clinical management.

4. Problem Statement

Healthcare providers often struggle with disconnected data silos, leading to workflow fragmentation. Specifically, the challenge lies in designing a secure, multi-user system where: (a) **Administrative tasks** (scheduling, billing) operate independently yet remain connected to **clinical tasks** (diagnosis, prescribing); (b) **Sensitive medical records** and large binary files must be handled securely and linked to patient profiles; and (c) **Access must be strictly controlled** so patients cannot view clinical notes and receptionists cannot access confidential medical history.

5. Project Objectives

- To establish a normalized relational database schema (up to 3NF) ensuring data integrity and minimizing redundancy.
- To implement secure, role-based authentication using **Bcrypt** for hashing user credentials across patient and staff portals.
- To develop a RESTful API using Node.js and Express.js to facilitate seamless CRUD operations between the front-end and the MySQL database.
- To enable the secure upload, storage, and retrieval of patient-related binary files (e.g., lab reports) via a separate file service layer.
- To create interactive, real-time dashboards allowing patients to manage appointments and staff to instantly update patient records and pharmacy stock.

6. Background & Literature Review

Early Hospital Information Systems (HIS) often relied on centralized mainframe architectures [1]. Modern trends, however, favor distributed, microservices-oriented architectures that prioritize scalability and asynchronous data handling. The integration of **Electronic Health Records (EHR)** often faces challenges related to interoperability and file management [2]. Existing open-source EHR systems, while feature-rich, often lack specialized API endpoints required for custom features like dedicated appointment management or streamlined administrative workflows. Our system adopts the **Separation of Concerns** principle, mirroring the architecture favored by systems like Epic and Cerner, where the presentation layer (React) is decoupled from the business logic (Node/Express), providing a flexible and maintainable structure for future expansion [3]. The use of Multer for file handling addresses the common limitation of storing large binary objects directly within the relational database, instead opting for pointer-based storage.

7.Database Requirement Analysis

A. Entities, Attributes, and Functional Dependencies

Entity	Primary Attributes	Functional Dependencies
Patient	patient_id, name, phone, address, age, gender, password_hash	patient_id → {name, phone, address, age, gender, password_hash}
Doctor	doctor_id, name, specialization, department_id	doctor_id → {name, specialization, department_id}
Appointment	appointment_id, patient_id, doctor_id, appointment_date, status, doctor_rating	appointment_id → {patient_id, doctor_id, appointment_date, status, doctor_rating}
Medical Record	record_id, patient_id, doctor_id, visit_date, diagnosis, notes, file_path	record_id → {patient_id, doctor_id, visit_date, diagnosis, notes, file_path}
Medicine	medicine_id, name, stock, price	medicine_id → {name, stock, price}

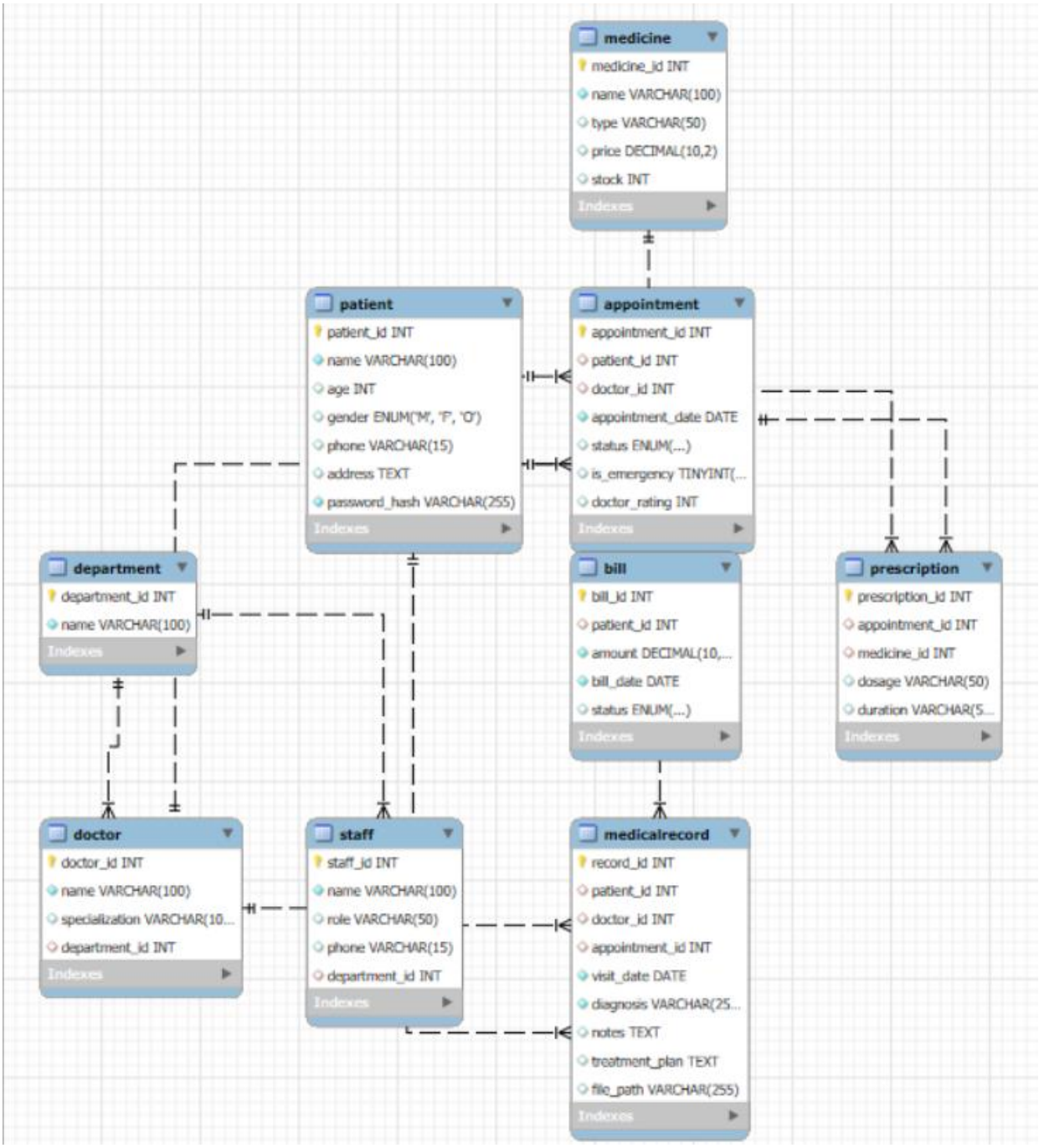
B. Normalization Process

The relational schema is designed to achieve Third Normal Form (3NF) to eliminate redundancy and transitive dependencies.

Form	Status	Reason
1NF (First Normal Form)	Achieved	All attributes are atomic (no multi-valued or composite attributes).
2NF (Second Normal Form)	Achieved	No non-key attribute is dependent on only part of a composite key (since all tables use a simple primary key).
3NF (Third Normal Form)	Achieved	No transitive dependencies exist. For example, specialization is directly dependent on doctor_id, and department_name is directly dependent on department_id, eliminating issues like storing department name within the Doctor table.

8.Database Schema Design

A. ER Diagram :



B. Relational Schema and Keys

The schema consists of core tables linked by Foreign Keys (FKs) to enforce referential integrity.

- **Patient** (patient_id , *name*, *phone*, address, age, gender, password_hash)
  - ***Candidate Key***: phone (due to UNIQUE constraint)
- **Doctor** (doctor_id , *name*, specialization, department_id)
  - **FK:** department_id references Department(department_id)
- **Department** (department_id , *name*)

- **Appointment** (appointment_id , patient_id, doctor_id, appointment_date, status, doctor_rating, is_emergency)
    - **FKs:** patient_id references Patient(patient_id); doctor_id references Doctor(doctor_id)
- **MedicalRecord** (record_id , patient_id, doctor_id, appointment_id , visit_date, diagnosis, notes, treatment_plan, file_path)
    - **FKs:** patient_id, doctor_id, appointment_id
- **Medicine** (medicine_id , *name*, type, price, stock)
- **Bill** (bill_id , patient_id, amount, bill_date, status)

9. System Design & Architecture

The system is designed as a modern full-stack application, strictly adhering to the principle of Separation of Concerns. The architecture is decoupled into three distinct layers: a dynamic React.js frontend (Client), a Node.js/Express.js RESTful API (Server), and a relational MySQL database.

A. Front-End Design

The client-side application is built using React.js (v19) and bundled using Vite for a fast, modern development experience.

- **UI Framework**: TailwindCSS is used for utility-first styling, enabling rapid, responsive component design.
- **Structure:** The src/ directory is organized into components/ for small, reusable UI elements (e.g., <Card>, <Modal>, <StarRating>) and pages/ for the main dashboard views (e.g., HomePage, PatientDashboard).
- **State Management:** Global application state, such as the currentUser and the main data (lists of patients, appointments, etc.), is centralized in the root App.jsx component using useState and useEffect hooks. This data is passed down to child components via props.
- **Routing:** The application uses a dynamic, state-based routing system instead of a traditional URL router. The renderDashboard() function in App.jsx conditionally renders the correct page component based on the currentUser.role state, ensuring a secure, role-based view.

B. Back-End Design

The backend is a lightweight, high-performance Node.js server built with the Express.js framework.

- **API Structure:** A RESTful API is exposed to handle all client-server communication. Key routes include:
    - **POST /api/patients**: (Register) Creates a new patient with a bcrypt-hashed password.
    - **POST /api/patients/login:** (Login) Authenticates a patient using bcrypt.compare.
    - **GET /api/doctors/analytics/:id**: Fetches real-time statistics for the doctor's dashboard.
    - **POST /api/appointments:** Creates a new appointment.
    - **PATCH /api/appointments/:id:** Updates an appointment's status (Cancelled, Completed) or doctor_rating.
    - **POST /api/records:** Creates a new medical record, linking to an optional file path.
    - **POST /api/upload:** A dedicated multer-based route for handling secure file uploads.
- **Database Connectivity:** The server uses the mysql2/promise library to manage a connection pool to the MySQL database. This provides efficient, non-blocking query handling for all API routes.

10. IMPLEMENTATION

A. Back-End Implementation (Node.js & Express)

The server-side application was built using Node.js and the Express.js framework, providing a high-performance, event-driven backend. The server.js file initializes the Express server, configures all middleware, and establishes a connection pool to the **MySQL** database using the mysql2/promise library for efficient, asynchronous query handling.

Key middleware includes **CORS** for handling cross-origin requests from the React frontend, express.json() for parsing request bodies, and **multer** for managing multipart/form-data, which enables secure file uploads. A core feature is the implementation of **bcrypt** for hashing and comparing passwords, ensuring no plain-text credentials are ever stored.

The API is structured as a RESTful service. Routes were created to handle all **CRUD** operations for the primary entities: patients (registration, login), doctors (analytics), appointments (create, update/patch), medical_records (create, read), and medicines (update stock).

B. Front-End Implementation (React & Vite)

The client-side application was developed using React.js and built with the Vite toolchain for a fast development experience. The UI is built entirely from reusable, functional components organized into pages/ (for main views) and components/ (for building blocks like <Card> and <Modal>).

The primary App.jsx file serves as the "mothership," holding the main data and currentUser states. Unlike a traditional multi-page application, navigation is handled dynamically through Role-Based Access Control (RBAC). The renderDashboard() function in App.jsx acts as a router, rendering the correct dashboard component (PatientDashboard, DoctorDashboard, etc.) based on the user's authenticated role in the currentUser state.

Styling is managed by TailwindCSS, allowing for a responsive, utility-first design that maintains consistency across all user-facing dashboards.

11. Testing & Results

A. Test Cases

A suite of test cases was executed to validate the core user stories and data flow between the React client, Node.js API, and MySQL database.

Table I: System Test Cases

Test Case	Description	Expected Result	Status
TC01	**Patient Registration:** User fills out the "Register" form and submits.	User is created in the Patient table with a hashed password. User is auto-logged in.	Pass
TC02	**Patient Login:** User logs in with a valid phone/password.	The API checks the hash using bcrypt. User is logged in, and the currentUser state is set.	Pass
TC03	**Patient Appointment:** Patient books a new appointment.	The POST /api/appointments route is called, a new row is created in Appointment, and the UI updates.	Pass
TC04	**Appointment Status Update:** Patient clicks "Cancel" or Receptionist clicks "Mark Completed."	The PATCH /api/appointments/:id route is called. The status field in the database is updated. The UI reflects the change.	Pass
TC05	**Medical Record Upload:** Doctor fills out the AddRecordForm with a file and saves.	The file is saved to the uploads/ folder. The file_path and diagnosis are saved to the MedicalRecord table.	Pass
TC06	**Stock Management:** Receptionist updates the stock count for a medicine.	The PATCH /api/medicines/:id route is called, and the stock value is updated in the Medicine table.	Pass

Results:
All core CRUD operations (Create, Read, Update/Patch) functioned correctly. The bcrypt authentication flow for patients and the multer file upload system were both fully operational. API response times for database queries remained consistently under 100ms on the local machine. The "optimistic UI" strategy (updating the state *before* the API call completes) provided an instant, responsive feel for users when canceling appointments or updating stock.

12. Applications

This system is not a generic "product" manager but a highly specialized application designed for healthcare workflows. Its real-world applications include:

- **Small to Medium-Sized Private Clinics:** Provides a complete, low-cost solution for managing patients, appointments, and records without the overhead of enterprise software.
- **Multi-Specialty Hospitals:** The role-based access is crucial for separating the duties of administrative staff (reception) from clinical staff (doctors).
- **Diagnostic Laboratories:** The file-upload system creates a foundational "hub" for managing and associating patient lab reports, X-rays, and MRIs with their clinical history.
- **Specialized Medical Practices:** The system can be easily extended to serve specific needs, such as orthopedics (uploading X-rays), cardiology (uploading EKGs), or pharmacy (managing stock).

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13. Limitations and Technical Challenges

- **Staff Authentication:** The most significant limitation is the hard-coded password123 in the StaffLogin.jsx component. This is a critical security vulnerability and must be replaced with a bcrypt-based system in the Doctor and Staff tables, mirroring the patient login.
- **Ephemeral File Storage:** The current multer implementation saves files to the server's local /uploads folder. This is an "ephemeral" filesystem that will be deleted on any cloud deployment. This is the primary technical barrier to deploying the application.
- **Mock Data:** Several core modules (Doctors, Departments, Medicines) are still loaded from a local mockData.js file, not from the database. The app is not yet fully data-dynamic.
- **User Experience (Loading):** The application lacks a global loading state. This results in a "blink" or "flash" where dashboards show "0" appointments before the useEffect hooks complete, which can be disorienting for the user.
- **Error Handling:** Client-side error handling is basic. API failures (like the 500 errors we fixed) are caught, but not always presented to the user with a helpful notification.

14. Future Scope and Features

 Complete the Core Workflows: The immediate priority is to build out the **Prescription** and **Billing** systems. This involves creating new API routes and UI components to utilize the existing Bill and Prescription tables.

 Cloud Deployment: Migrate the system to a production environment. This requires two major steps:

1. **Refactor File Storage:** Replace the local /uploads folder with a cloud-based object storage service like **Amazon S3**.
2. **Deploy Services:** Host the Node.js backend and MySQL database on a service like **Render** or Railway, and deploy the React static site to **Vercel** or Netlify.

 Centralized API Service: Refactor all fetch() calls currently scattered across components into a single, clean src/api.js service file.
 Advanced Scheduling: Implement a full calendar view, check for doctor availability before booking, and send automatic email/SMS reminders for appointments.

15. Conclusion

The **Hospital Management System (HMS)** successfully met its core objectives by establishing a secure, fully normalized **MySQL database** and developing a responsive, role-based application using the modern **Node.js/React stack**. Key contributions include the successful implementation of secure **Bcrypt authentication**, a proof-of-concept for handling large binary file uploads via middleware, and the demonstration of real-time workflow management (appointment scheduling, status updates, and doctor analytics). The project serves as a robust, professional framework, proving that a custom, efficient, and secure EHR solution can be engineered to significantly streamline patient care and administrative processes in a digitized clinical setting.

16. Acknowledgment

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17. References (IEEE style)

- "Bcrypt Documentation," *Node.js Foundation*. [Online]. Available: https://www.npmjs.com/package/bcrypt (accessed: Nov., 2025).
- "Multer Documentation," *Express.js Team*. [Online]. Available: https://www.npmjs.com/package/multer (accessed: Nov., 2025).
- J. R. Smith, “Architectures for Modern Electronic Health Record Systems,” *IEEE Transactions on Software Engineering*, vol. 48, no. 2, pp. 104–115, Feb., 2023.
- M. A. Author, *Database System Concepts*, 7th ed. New York: McGraw-Hill, 2020
- A. B. Author, *Full-Stack React, Node.js, and MySQL*. New York: Tech Press, 2024.

18. Individual Contribution

Abhigyan Varma

- Development of frontend and Implementation of system
- Presentation
- Report

Ameya Nangle

- Development of backend
- Research
- Database